Experiment -5.1

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- **Aim:**
  To develop and configure a Spring Boot application that exposes REST APIs, serves HTML views, implements basic Spring Security, and to design an Entity-Relationship diagram for a healthcare application.

- **Objectives:**
  To create a basic Spring Boot project with a GET API returning a JSON response. To create a GET API that resolves and returns an HTML view. To secure the HTML view using Spring Boot Security. To design and visualize the database schema for a Healthhub project using an ER diagram

- **Implementation:**

  1. First, initialize a Spring Boot application using Spring Initializr with dependencies for Spring Web, Spring Thymeleaf, and Spring Security.
  2. Next, create a REST Controller class annotated with RestController to handle HTTP GET requests and return data in JSON format.
  3. Then, create a standard Controller class annotated with Controller to map a URL endpoint to an HTML template located in the resources templates directory.
  4. After that, configure Spring Security by adding the starter security dependency, which automatically secures all endpoints, and define user credentials in the application properties file.
  5. Finally, identify the core entities of the Healthhub project, such as Patient, Doctor, Appointment, and Medical Record, and map their relationships to generate an ER diagram.

- **Output:**

r.java | ApiController.java | ViewCont

src > main > java > com > example > healthHub > controllers > ApiController.java > ...

```java
package com.example.healthHub.controllers;

import java.util.*;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.RequestMapping;
import org.springframework.web.bind.annotation.RestController;

@RestController
@RequestMapping("/api")
public class ApiController {
  @GetMapping("status")
  public Map<String,String> getSystemStatus(){
    Map<String,String> mp=new HashMap<>();
      mp.put("status","Success");
      mp.put("message","HealthHub API is up and running ");
      return mp;
  }
}
```

localhost:8080/api/status

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Pretty-print

```
{
  "message": "HealthHub API is up and running ",
  "status": "Success"
}
```

src > main > java > com > example > healthHub > controllers > ViewController.java > ...

```java
package com.example.healthHub.controllers;

import org.springframework.stereotype.Controller;
import org.springframework.ui.Model;
import org.springframework.web.bind.annotation.GetMapping;

@Controller
public class ViewController {
  @GetMapping("/home")
  public String getHomePage(Model model){
    model.addAttribute("title","Welcome to Healthhub Dashboard");
    return "index";
  }
}
```

localhost:8080/home?continue

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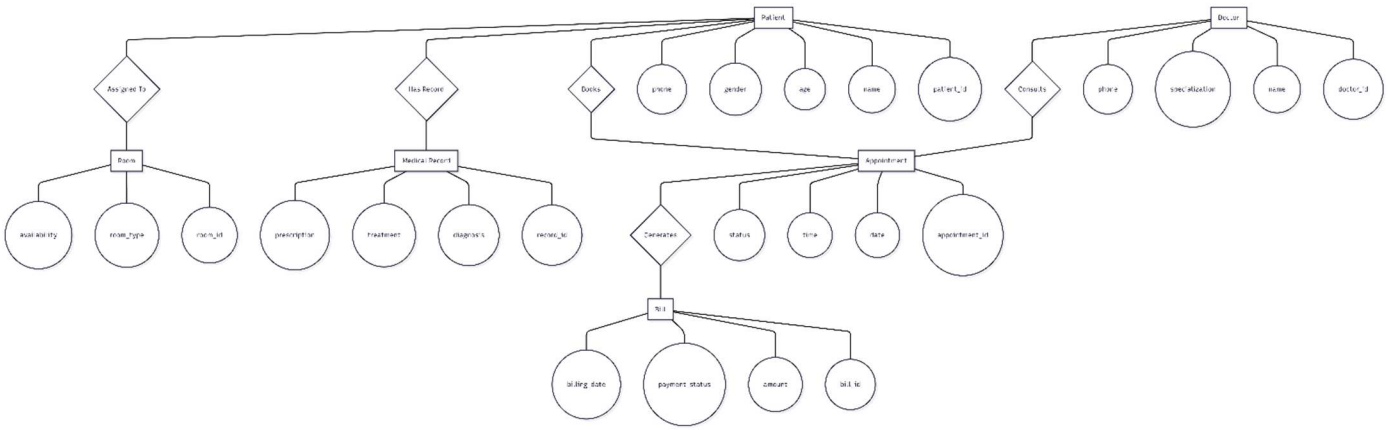
Welcome to Healthhub Dashboard

If you are seeing this, you have successfully logged in via Spring Security!



- **Results:**

  A functional Spring Boot application was successfully built. It returns JSON data, renders HTML views, and restricts access via Spring Security. Additionally, the database structure for the Healthhub system was mapped out.

- **Learning Outcomes:**
  1. Understood the fundamental differences between RestController and Controller in Spring Boot.
  2. Learned how to serve static and dynamic web content.
  3. Gained practical experience in applying basic authentication using Spring Security.
  4. Developed database design skills by creating an ER diagram for a specific domain.